

and quick solution times. Analysts need to determine the time available for the initial optimization as well as real-time re-optimizations (described below) to determine the best approach given time constraints.

PORTFOLIO ADAPTATION TACTICS

The optimization techniques provided above provide investors and algorithms with appropriate initial trading strategies. These trade schedules were determined based on expected market conditions and price movement. Unfortunately, the only thing we are sure about with regards to markets is that actual conditions will not be the same as expected conditions.

To adjust for changing conditions during the day investors can utilize the same adaptation tactics with a portfolio as they can with single stock trading (see Chapter 8). In this section we discuss the AIM and PIM tactics for portfolio trading needs.

The AIM and PIM tactics are:

$$AIM: \quad \text{Max} \quad \frac{E_t[Cost] - E_0[Cost]}{E_t[Timing Risk]} \quad (9.17)$$

$$PIM: \quad \text{Max} \quad \frac{E_0[Cost] - E_t[Cost]}{E_t[Timing Risk]} \quad (9.18)$$

where,

$E_0[Cost] = C^* = \text{original estimated cost (including permanent impact)}$

$E_t[Cost] = \text{expected total cost at time } t \text{ (includes realized and unrealized)}$

$E_t[Timing Risk] = \text{expected timing risk at time } t \text{ (unexecuted shares only)}$

The original cost estimate is determined from the original optimization solution. It includes temporary impact, permanent impact, and price appreciation. Even if the optimization does not include the permanent impact component (for optimization simplification) the permanent impact cost needs to be added into the estimated cost. Permanent impact cost is a true cost to investors but since it will not influence the optimization solution it is often not included in the optimization formulation.

The time expectations for cost and timing risk are computed as follows:

Then our cost equations are:

$$Realized_s(Cost(t)) = \sum_i Side(i) \cdot Y_i \cdot (P_{ij} - P_{i0}) \quad (9.19)$$